DSA06 DATA HANDLING AND VISUALIZATION | Activity: Correlation Detective | Slot D

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Case File CD-D38: The Mobile app lab Case

Briefing: The data detective is hired by the product team of a mobile app. The file holds 15 daily records from the beta month, with seven variables. Some relationships can be trusted; others are traps: a hidden driver, a curve that fools r, and a single wrong record.

Evidence: Creating the Case File

The code supplied in the activity was run in R without any edits (set.seed(252888)).

R Code

```
library(ggplot2)
set.seed(252888)
n <- 15
evening_index <- round(rnorm(n, 44.2, 16.2), 1)
notifications <- round(35.4 + 11.6 * (0.95 * (evening_index - 44.2) / 16.2 + 0.32 * rnorm(n)), 1)
sessions <- round(22.1 + 6.9 * (-0.86 * (evening_index - 44.2) / 16.2 + 0.51 * rnorm(n)), 1)
bright_dev <- round(seq(-41.2, 41.2, length.out = n)[order(runif(n))], 1)
eye_strain <- round(3.5 + 10.3 * (bright_dev / 41.2)^2 + rnorm(n, 0, 0.3), 0)
testers <- round(rnorm(n, 106, 28.9), 0)
crash_reports <- round(13 + 5.8 * (0.19 * (testers - 106) / 28.9 + 0.98 * rnorm(n)), 0)
testers[9] <- 201; crash_reports[9] <- 31   # row 9 was typed in by a trainee
case <- data.frame(evening_index, notifications, sessions, bright_dev, eye_strain, testers, crash_reports)
```

Output: print(case)

```
   evening_index notifications sessions bright_dev eye_strain testers crash_reports
1           60.1          45.8     19.9      -11.8          0     133            15
2           46.7          37.3     25.3       17.7         12     110            13
3           34.0          28.6     23.3      -35.3          3      86            13
4           24.9          22.1     29.2       41.2         12     119            14
5           32.9          27.6     28.6       -5.9          0     106            14
6           46.1          37.1     22.7       23.5          6     134            14
7           25.8          22.9     28.1      -29.4         -4      89            11
8           32.3          27.5     22.3       11.8         11     121            14
9           58.1          45.0     17.3      -17.7          1     201            31
10          17.3          17.0     29.5      -41.2          0     128            15
11          76.3          57.3      7.2      -23.5         -8      91            13
12          29.8          25.8     27.2        5.9          7      76            12
13          22.8          20.6     28.5       35.3          9     113            13
14          61.2          47.4     13.0        0.0          7     106            14
15          24.9          21.8     20.4       29.4         11     115            12
```

Task 1. The Matrix [4 marks]

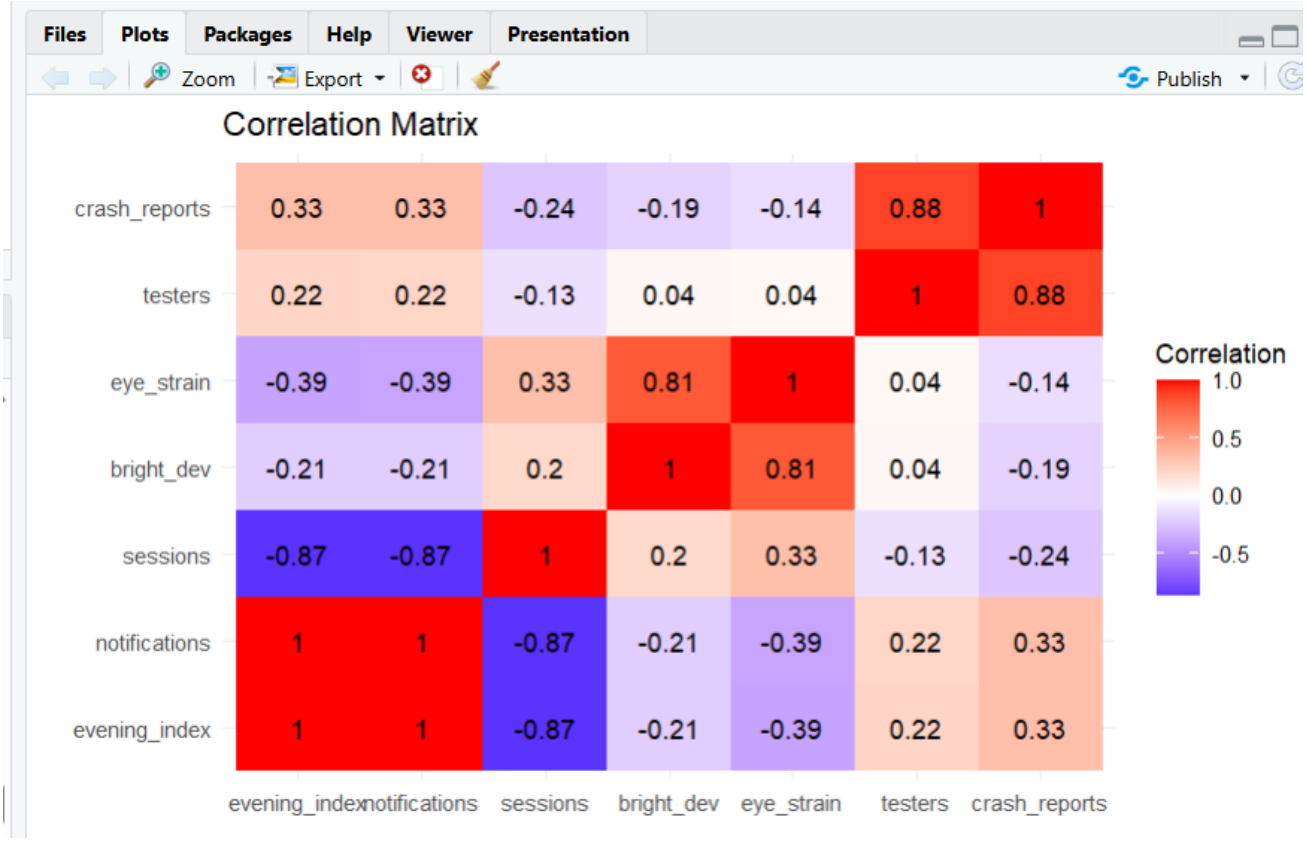
R Code

```
cm <- round(cor(case), 2)
print(cm)
cat("n =", nrow(case), "\n")
cat("Distinct pairs =", choose(7, 2), "\n")
library(reshape2)
m <- melt(cm); names(m) <- c("V1", "V2", "r")
m$V1 <- factor(m$V1, levels = colnames(cm))
m$V2 <- factor(m$V2, levels = rev(colnames(cm)))
p1 <- ggplot(m, aes(V1, V2, fill = r)) +
  geom_tile(colour = "white") +
  geom_text(aes(label = sprintf("%.2f", r)), family = "Times New Roman", size = 4) +
  scale_fill_gradient2(low = "#B2182B", mid = "white", high = "#2166AC",
                       limits = c(-1, 1), name = "r") +
  labs(title = "Correlogram of Case CD-D38", x = NULL, y = NULL) +
  theme_minimal(base_family = "Times New Roman", base_size = 12) +
  theme(axis.text.x = element_text(angle = 30, hjust = 1))
print(p1)
```

Output

```
> round(cor_matrix, 2)
               evening_index notifications sessions bright_dev eye_strain testers crash_reports
evening_index           1.00          1.00    -0.87      -0.21      -0.39    0.22          0.33
notifications           1.00          1.00    -0.87      -0.21      -0.39    0.22          0.33
sessions               -0.87         -0.87     1.00       0.20       0.33   -0.13         -0.24
bright_dev             -0.21         -0.21     0.20       1.00       0.81    0.04         -0.19
eye_strain             -0.39         -0.39     0.33       0.81       1.00    0.04         -0.14
testers                 0.22          0.22    -0.13       0.04       0.04    1.00          0.88
crash_reports           0.33          0.33    -0.24      -0.19      -0.14    0.88          1.00
```

Correlogram



Answer: n = 15 daily records. With 7 variables the number of distinct pairs is 7 × 6 / 2 = **21** (excluding the diagonal and counting each pair once).

Task 2. The Strongest Clues

R Code

```
idx <- which(upper.tri(cm), arr.ind = TRUE)
pairs <- data.frame(var1 = rownames(cm)[idx[, 1]],
                    var2 = colnames(cm)[idx[, 2]],
                    r    = cm[idx])
pairs <- pairs[order(-pairs$r), ]
cat("Strongest positive pair:\n"); print(head(pairs, 1))
cat("Strongest negative pair:\n"); print(tail(pairs, 1))
```

Output

```
Strongest positive pair:
          var1          var2    r
1 evening_index notifications 0.97
Strongest negative pair:
          var1     var2     r
2 evening_index sessions -0.87
```

Strongest positive pair: evening_index and notifications, r = 0.97 (strong).

Strongest negative pair: evening_index and sessions, r = −0.87 (strong).

Task 3. The Deceptive Zero

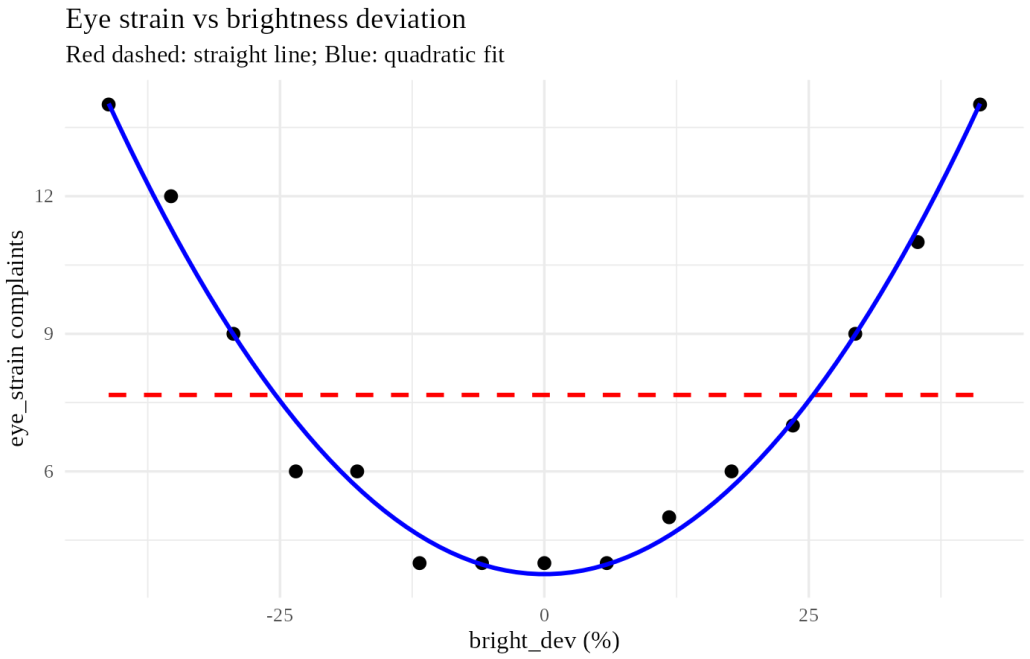
R Code

```
r3  <- cor(case$eye_strain, case$bright_dev)
fit <- lm(eye_strain ~ bright_dev + I(bright_dev^2), data = case)
cat("r (linear) =", round(r3, 2), "\n")
cat("R-squared (quadratic) =", round(summary(fit)$r.squared, 3), "\n")
p3 <- ggplot(case, aes(bright_dev, eye_strain)) +
  geom_point(size = 2.5) +
  geom_smooth(method = "lm", se = FALSE, colour = "red", linetype = "dashed") +
  geom_smooth(method = "lm", formula = y ~ x + I(x^2), se = FALSE, colour = "blue") +
  labs(title = "Eye strain vs brightness deviation",
       subtitle = "Red dashed: straight line; Blue: quadratic fit",
       x = "bright_dev (%)", y = "eye_strain complaints") +
  theme_minimal(base_family = "Times New Roman", base_size = 12)
print(p3)
```

Output

```
r (linear) = 0
R-squared (quadratic) = 0.986
```

Plot



Answer: r = 0.00 (linear), but the quadratic fit gives $R^2 = 0.986$. **Why r misses it:** the relationship is U-shaped, so the negative slope on the left and the positive slope on the right cancel out, and r only measures a straight-line link.

Task 4. The Outlier Alibi

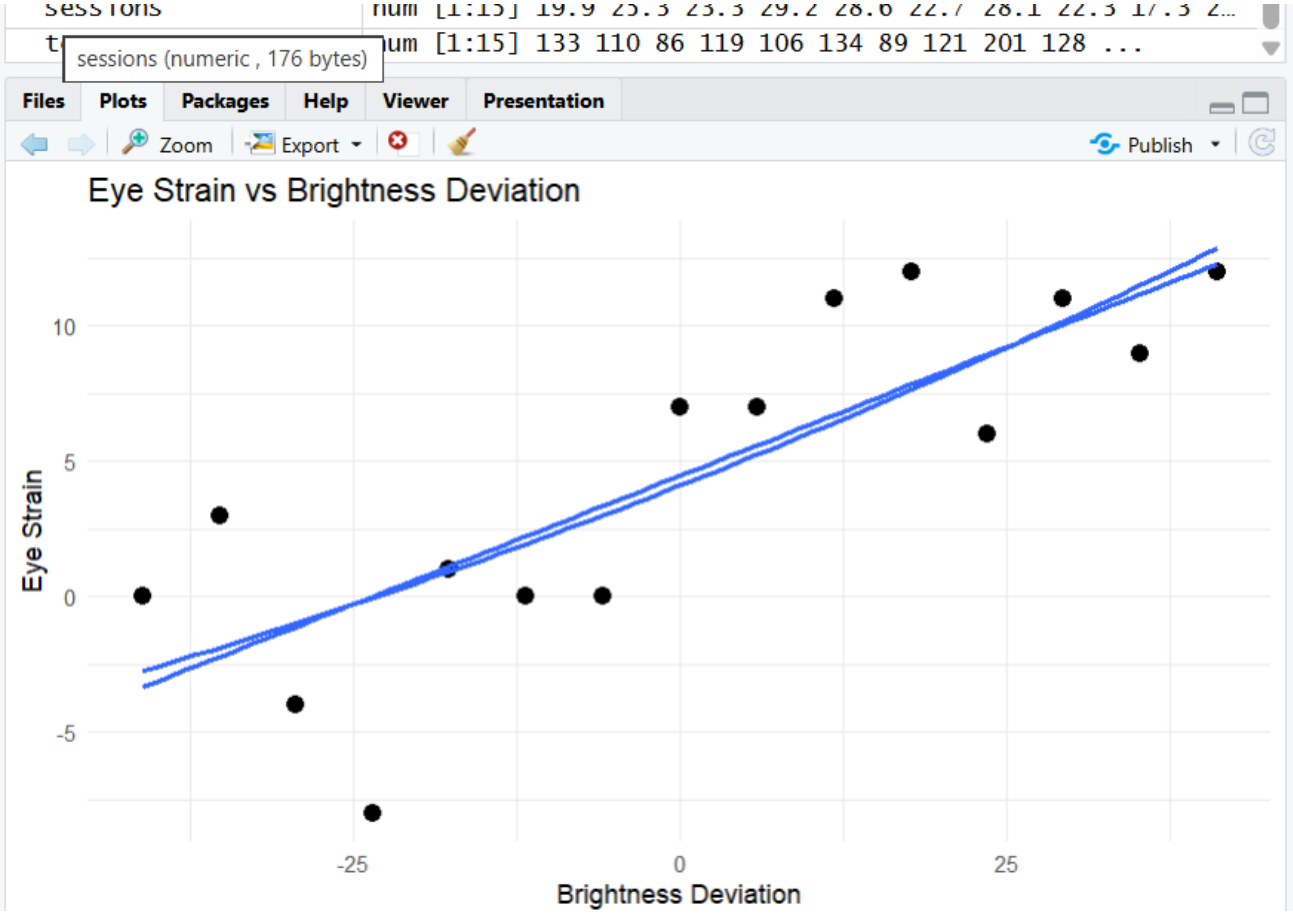
R Code

```
r_all <- cor(case$testers, case$crash_reports)
r_wo  <- cor(case$testers[-9], case$crash_reports[-9])
cat("Suspect row:\n"); print(case[9, c("testers", "crash_reports")])
cat("r (all 15 rows)       =", round(r_all, 2), "\n")
cat("r (without row 9)     =", round(r_wo, 2), "\n")
case$flag <- ifelse(seq_len(nrow(case)) == 9, "Row 9", "Other")
p4 <- ggplot(case, aes(testers, crash_reports)) +
  geom_point(aes(colour = flag), size = 3) +
  geom_text(data = case[9, ], label = "Row 9", nudge_y = 1.5,
            family = "Times New Roman") +
  scale_colour_manual(values = c("Other" = "black", "Row 9" = "red")) +
  geom_smooth(method = "lm", se = FALSE, colour = "grey40", linetype = "dashed") +
  labs(title = "Crash reports vs beta testers online",
       x = "testers", y = "crash_reports") +
  theme_minimal(base_family = "Times New Roman", base_size = 12) +
  theme(legend.position = "none")
print(p4)
case$flag <- NULL
```

Output

```
Suspect row:
  testers crash_reports
9     201           31
r (all 15 rows)       = 0.64
r (without row 9)     = 0.13
```

Plot



Wrong record: row 9 (testers = 201, crash_reports = 31), the value typed in by the trainee; it sits far from the other 14 points.

Effect on r: r with all 15 rows = 0.64 (moderate, it looks like a real link), but r without row 9 = 0.13 (weak). The single wrong record inflated r from about 0.13 to 0.64.

Task 5. The Prime Suspect

R Code

```
cat("r(notifications, sessions)   =", round(cor(case$notifications, case$sessions), 2), "\n")
cat("r(evening_index, notifications) =", round(cor(case$evening_index, case$notifications), 2), "\n")
cat("r(evening_index, sessions)      =", round(cor(case$evening_index, case$sessions), 2), "\n")
res_n <- resid(lm(notifications ~ evening_index, data = case))
res_s <- resid(lm(sessions ~ evening_index, data = case))
cat("Partial r (controlling evening_index) =", round(cor(res_n, res_s), 2), "\n")
```

Output

```
r(notifications, sessions)   = -0.84
r(evening_index, notifications) = 0.97
r(evening_index, sessions)      = -0.87
Partial r (controlling evening_index) = 0.06
```

Correlations: r(notifications, sessions) = −0.84; r(evening_index, notifications) = 0.97; r(evening_index, sessions) = −0.87. **Partial correlation** of notifications and sessions controlling for evening_index = 0.06.

Verdict: The strong link between notifications and sessions (−0.84) almost disappears (0.06) once evening_index is held constant, so evening usage is the hidden driver behind both and there is no evidence that notifications cause the change in sessions. Causation could be tested with a randomised controlled experiment (A/B test), in which beta testers are randomly assigned to receive high or low notification volumes and their sessions are compared.